Is there a significant correlation between TA population size and theft rate, and well can population predict theft rate (per 1,000 residents) across NZ TAs, FY2023–FY2025?

DATA 301 Phase 2 Report

Trimester 2, 2026

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Executive summary

This project explored recorded victimisations in New Zealand across three financial years: 2023, 2024, and 2025. The recorded victimisations were population-adjusted and calculated across Territorial Authorities (TAs) in New Zealand. A key limitation of these victimisation rates is that they are based on recorded crime rather than all actual crime, which fails to account for unreported or unrecorded offenses.

I observed large variability in victimisation rates across TAs. Across all offence types, Hamilton City had the highest recorded victimisation rate at 105.25 per 1,000 residents, while Central Otago District had the lowest at 10.74 per 1,000 residents. The largest offence category was theft, accounting for 67.73% of total recorded victimisations across the six categories. I focused my Phase 2 research on this category to explore whether there was a significant correlation between theft rate and TA population size, helping us understand whether theft rates rise as population size increases.

The correlation analysis revealed a moderate positive relationship (correlation value = 0.63) between recorded theft rates and TA population size, indicating that larger TAs tend to experience higher theft rates. The 95% confidence interval [0.52, 0.70] indicates we can be 95% confident that the true population correlation lies within this moderate-to-strong positive range. This correlation remained consistent when analyzing individual years (ranging from 0.60 to 0.65, all $p < 0.001$) confirming that the finding is stable over time.

I further investigated how effectively population size could predict theft rates. The model tested had a typical miss of 12.1 units of recorded thefts per 1,000 residents, compared to a miss of 15.6 for a simple baseline guess (measured in Root Mean Square Error/RMSE). I found that using population improved predictions, achieving an RMSE reduction of 2.83 units compared to the baseline model, but it left considerable variation unexplained ($R^2 = 0.363$/ ~ two thirds of data unexplained) demonstrating that population size alone is not a strong single predictor.

1. Background

Recorded crime by area is important because it allows government authorities to shift the focus from "Who's committing the crime?" to 'Where are crimes happening?". This will support the government to make informed decisions to allocate resources equitably and provide different TAs in New Zealand with the support they need (New Zealand Police, n.d.). This research focuses on the victimisations that occur in TAs in New Zealand, using victimisation data from the New Zealand Police and population data from Stats NZ. New Zealand Police records crimes in TAs which are areas that are defined by its legal and geographical boundaries. This information is also wanted by many new Zealanders who are interested in knowing how their neighbourhoods differ in comparison to others (Stats NZ, 2013).

In Phase 1 of our project, we found that theft was the biggest category of crime, making up about 68% of all recorded crime in the data. But then we found that theft rates varied a lot between areas, from about 10 per 1,000 people in Central Otago, up to 80 per 1,000 in Hamilton. This led me to question whether population size could explain that difference in theft rates. Which is why I shifted my focus of the research to the theft rate of TAs to assess if there are correlations between the TAs population size and theft rate. I further investigated how well the population can predict the theft rate. The theft rate I used was the recorded theft count from the victimisation data divided by its TA value and scaled to 1,000 residents. This makes it population adjusted allowing us to use it for our analysis.

2. Data Description

2.1 Datasets

Dataset #1: NZ Police, Victimisation Time and Place.

The police victimisation data was sourced from the official New Zealand Police website which give us the Recorded Crime Victims Statistics (RCVS), Victimisation Time and Place report (New Zealand Police, n.d.). This victimisation data measures the counts of recorded victimisations by offence type under the Australian and New Zealand Standard Offence Classification (ANZSOC) Division, month, and location, including TA and Area Unit. This dataset combines offence type, time, and geographic information, making it suitable for comparing recorded victimisation patterns across TAs and over time. This is useful for understanding how recorded victimisation counts vary across TAs and over time; which offence types are more common in different TAs; and how theft rate differs across TAs.

The dataset is a long-format table, with 891,389 rows and 5 fields:

Field	Type	Example
ANZSOC Division	Categorical (text)	"Burglary", "Assault", "Theft"
Year Month	Date (month year)	"July 2022"
Victimisations	Numeric (count)	1,2,3, ...
Territorial Authority	Categorical (text)	"Auckland", "Porirua City"
Area Unit	Categorical (text)	"Boulcott", "Taihape"

Table 1: Structure of police victimisation data (New Zealand Police, n.d.).

The analysis uses data from July 2022 to June 2025 (36 months), which are grouped into three year-ended-June periods to align with the Stats NZ population estimates. In the cleaned analysis data, no null values were found, and all 67 Territorial Authorities are represented across each of the three financial years. Six offence categories are present. In the offence-level dataset, 72 TA-year-offence combinations with zero recorded victimisations were represented by absent rows; these were added explicitly as zero-count observations in the analysis-ready dataset without changing any victimisation totals.

Dataset #2: Stats NZ – Subnational Population Estimates (Territorial Authority).

This Stat NZ population data was sourced from Stats NZ to show us the Subnational Population Estimates, and Estimated Resident Population series (Stats NZ, 2025). The population data measures the annual population estimate per TA, on 30 June each year. This is important because the raw victimisation counts are strongly influenced by population size. This dataset provides the population denominator needed to calculate victimisations per 1,000 residents, allowing more meaningful comparisons between TAs of different sizes. It also shows how TA populations change over time and, when combined with victimisation data, how recorded victimisation rates per 1,000 residents vary across TAs and financial years.

The structure of this dataset is a wide-format table with one row per year and one column per Territorial Authority for 2015–2025; values are integer population counts. This dataset has no missing or suppressed values. The source population data cover 2015–2025, but only the 2023–2025 estimates are used to match the three financial-year periods in the victimisation data.

2.2 Integration steps

The following steps ensured the datasets were cleaned and ready for analysis. First the population data was reshaped from wide to long format (Year, TA, Population). This was to allow the libraries and tools to easily identify the columns and its values because tools such as Python and R code work best when datasets are in long format. TA names were also standardised to align in both datasets ("Wanganui District" -> "Whanganui District"). Victimisation data was also grouped into financial years (FY2023 = July 2022–June 2023, FY2024 = July 2023–June 2024, and FY2025 = July 2024–June 2025). The previous two steps ensured the datasets column align with each other so we could use both to compute the different tests. The merged data was later filtered to victimisation type ‘theft’, this will be important since our second phase of our report will focus on this. The victimisation data was joined with the population data and included theft rate which took the recorded theft count from the victimisation data divided by its TA value and scaled to 1,000 residents. A log function was also applied to population because the data was skewed and had a lot of variation, which this function was able to reduce.

3. Ethics, Privacy and Security

3.1 Ethical Considerations

An ethical consideration relates to how conclusions from this data could be perceived and used by the public. If the analysis shows that a particular Territorial Authority has a higher recorded victimisation rate than others, readers may interpret that area as inherently less safe even though the recorded rate does not necessarily reflect an individual’s actual risk of victimisation. Presenting these differences without sufficient context could stigmatise particular communities or encourage readers to treat recorded victimisation rates as direct measures of how dangerous an area is. Recorded victimisation data reflect offences that have been reported to and recorded by Police, and these counts may be influenced by differences in reporting behaviour or policing activity rather than directly representing individuals’ underlying risk of victimisation. Differences in reporting behaviour and policing activity may therefore affect recorded counts across Territorial Authorities, so these figures should not be interpreted as direct measures of underlying victimisation risk. Our group therefore presents comparisons between Territorial Authorities as differences in recorded victimisation rates rather than as a simple ranking of "dangerous" and "safe" areas.

3.2 Privacy Concerns

There are several privacy considerations associated with the Police data, particularly because the source data include geographic information that could increase disclosure risk in small or low-count areas. The Victimisation Time and Place report removes sensitive details at the detailed time-and-place level, including victimisations occurring in dwellings other than burglary, to protect the privacy of individuals (New Zealand Police, n.d.). This supports the confidentiality principle of the CIA triad by reducing the risk that published location information could be linked to a specific victim or private residence. Even without direct identifiers, combining detailed geographic information with external local knowledge could increase the risk of identifying a victim or property associated with a recorded incident. This is a particular risk for crimes at a private residence, since a home address is a much stronger identifier than a crime occurring in a public place.

A further privacy consideration is that the Police source includes geographic information at Area Unit level (New Zealand Police, n.d.). For crime types with a low number of victimisations in a particular area unit, this could arguably make it easier for someone to connect a specific incident to a specific person or property, especially if they already have some local knowledge of the area. Area Unit provides a finer level of geographic detail than Territorial Authority, which can increase disclosure risk for low-count observations when combined with local knowledge. Although the source Police data contain finer geographic information, including Area Unit, this project aggregates and reports results at Territorial Authority level, which reduces the risk of drawing attention to individual low-count local areas while retaining the geographic detail needed for the analysis.

3.3 Data and Results Security

There are certain steps our group has taken to ensure the security of the data and results. Our GitHub repository, which includes the dataset and analysis code, has been made private and is only accessible to group members. The repository does not contain any data broken down more finely than what is already publicly available through the Police and Stats NZ sources, and our code does not produce or store any output at a finer geographic level than Territorial Authority. We used the GitHub repository as the central location for shared project files and maintain clearly named versions of the cleaned and analysis-ready datasets, reducing the risk of group members using outdated or inconsistent files. With additional time and resources, we would formalise these practices through a documented data-management plan, periodically review repository access, and archive or remove unnecessary working copies once the project is complete.

4. Exploratory Data Analysis

4.1 National Recorded Victimisations by Offence Category (FY2023–FY2025)

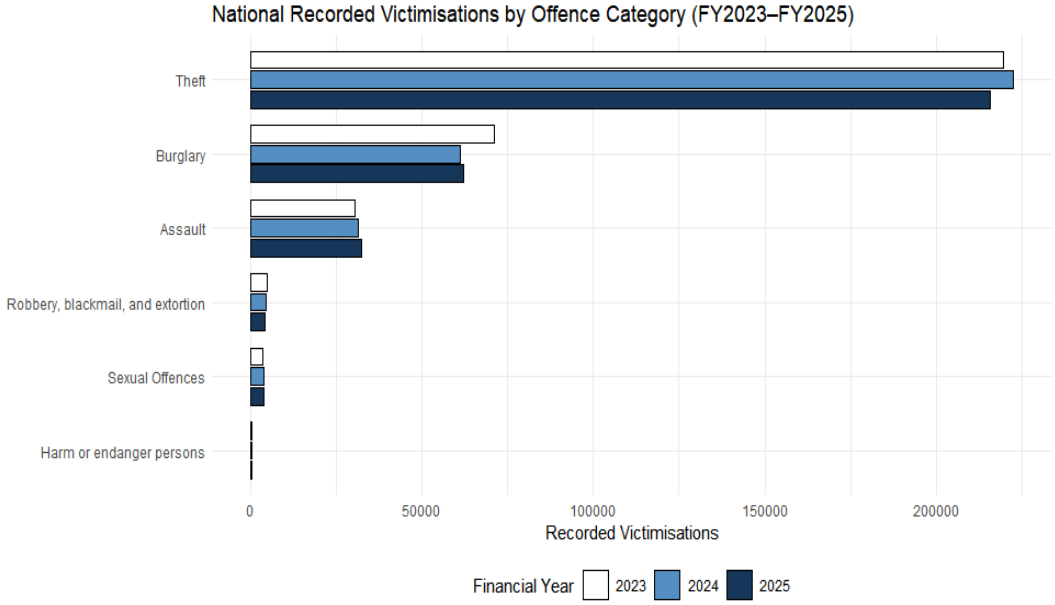


Figure 1. Total recorded victimisations across New Zealand broken down by ANZSOC Division category for FY2023, FY2024, and FY2025.

Figure 1 shows that property-related recorded victimisations dominated the six offence categories analysed, with Theft accounting for 67.73% (658,029) and Burglary accounting for 19.99% (194,205)

of recorded victimisations over the three-year period. Combined, Theft and Burglary accounted for 87.72% of recorded victimisations across the six offence categories analysed. The theft variable remained the single largest offence type, peaking in FY2024 at 222,652 victimisations (42.091 per 1,000 residents) before easing to 215,637 in FY2025 (40.500 per 1,000). Burglary decreased substantially after FY2023, from 71,027 recorded victimisations to 61,154 in FY2024, and remained relatively stable in FY2025 at 62,024. For Assault, recorded victimisations increased each year, from 30,486 in FY2023 to 32,178 in FY2025, a 5.55% increase in counts; the corresponding national rate rose from 5.863 to 6.043 per 1,000 residents. Other offences such as Robbery, blackmail, and extortion decreased from 4,663 recorded victimisations in FY2023 to 4,217 in FY2025, while Sexual Offences increased slightly from 3,552 to 3,760. Harm or endanger persons accounted for less than 0.1% of recorded victimisations across the six offence categories analysed.

4.2 FY2025 Recorded Victimisation Rates Across Territorial Authorities

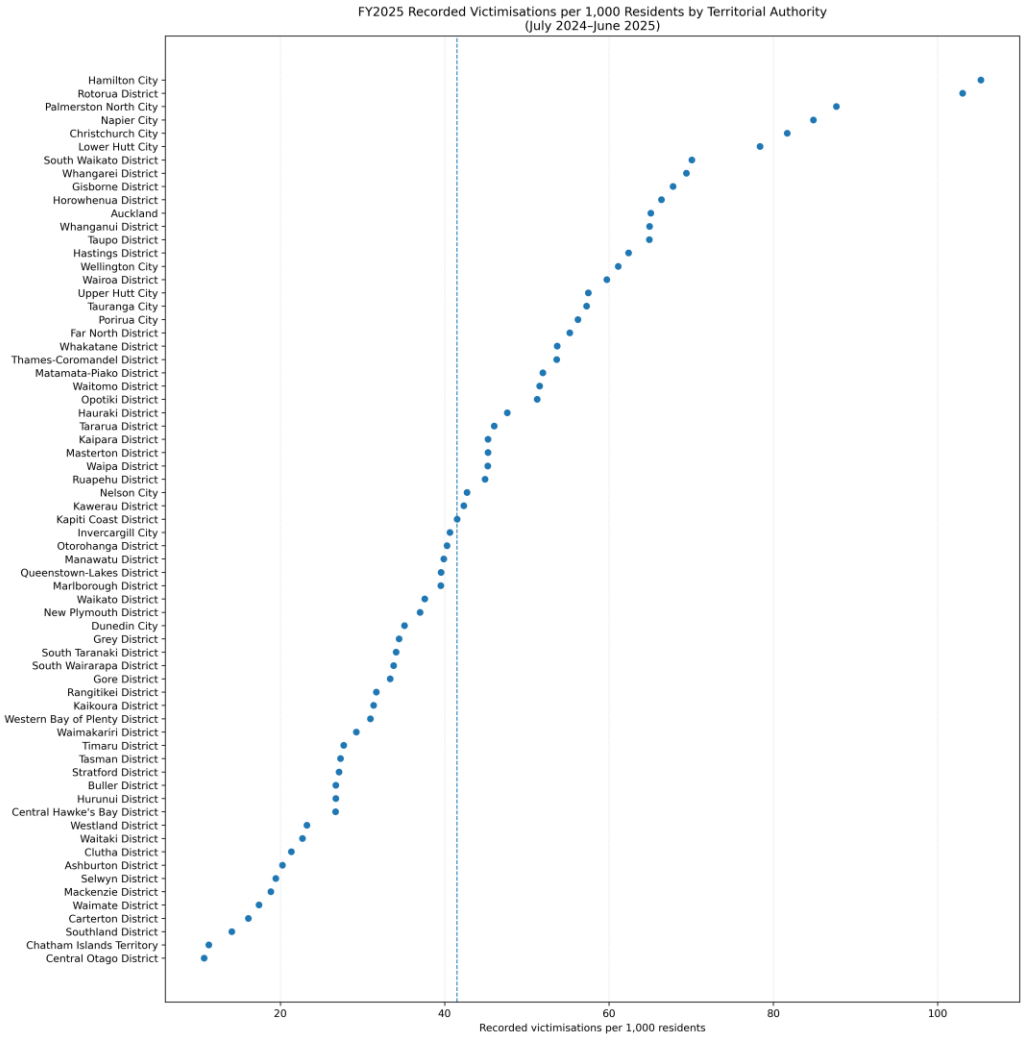


Figure 2. Recorded victimisation rates per 1,000 residents across New Zealand Territorial Authorities in FY2025 (July 2024–June 2025). Territorial Authorities are ordered by recorded victimisation rate. The dashed vertical line indicates the median rate of 41.517 per 1,000 residents.

Figure 2 shows that recorded victimisation rates varied substantially across Territorial Authorities in FY2025. Hamilton City had the highest rate at 105.252 victimisations per 1,000 residents, followed closely by Rotorua District at 103.026. In contrast, Central Otago District recorded the lowest rate at 10.736 per 1,000. The median rate across the 67 Territorial Authorities was 41.517 per 1,000, indicating considerable geographic variation in recorded victimisation relative to resident population.

4.3 Queenstown-Lakes District: An Increase Against the National Trend

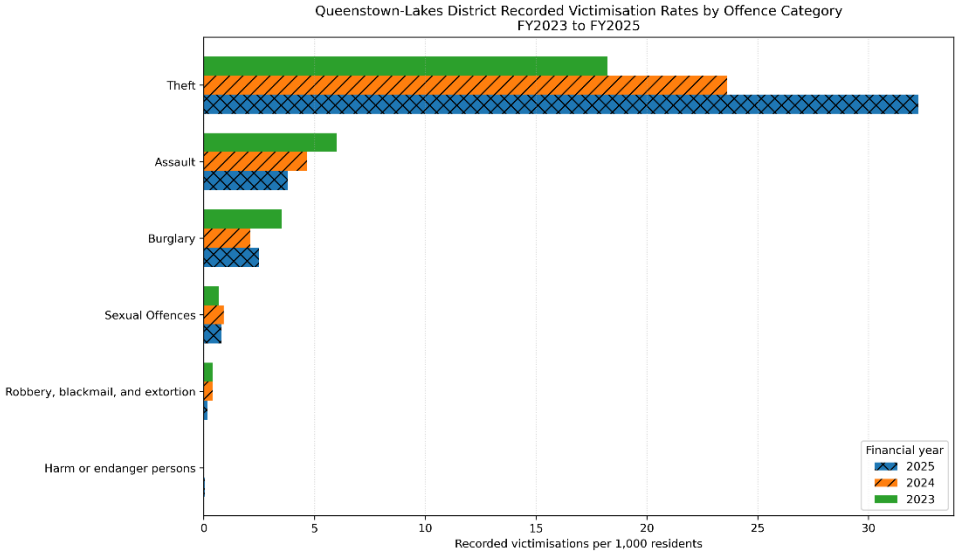


Figure 3. Recorded victimisation rates per 1,000 residents in Queenstown-Lakes District by offence category for FY2023, FY2024 and FY2025.

Figure 3 compares the different victimisations in Queenstown-Lakes District, and that the increase was concentrated primarily in theft. The recorded theft rate rose from 18.214 per 1,000 residents in FY2023 to 23.629 in FY2024 and 32.249 in FY2025, an overall increase of 14.035 per 1,000. By comparison, assault declined from 6.000 to 3.792 per 1,000 and burglary declined from 3.534 to 2.491. Changes in the remaining offence categories were comparatively small. Therefore, the rise in Queenstown-Lakes' overall recorded victimisation rate was principally accounted for by increased recorded theft victimisations, while declines in several other categories partially offset this increase.

4.4 Theft Rate Against Log Population

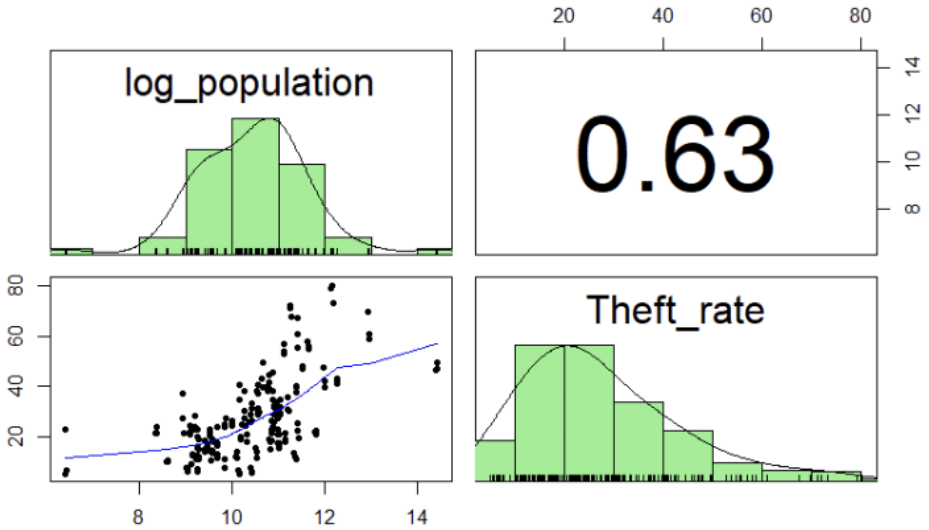


Figure 4. Spearman pair.panel plot of variables; log(population) and theft rate per 1,000 residents. The recorded theft rate points (201) are 67 TA points across three years (FY23-FY25).

Figure 4 shows the log population against recorded theft rates, which have values for each year (FY23, 24, 25). The Spearman test will treat each of the 201 values as independent, which is each TA once per year. The histograms show that log(population) is roughly symmetrical, but theft rate is right-skewed. The correlation value of 0.63 shows a moderate correlation between the two variables, which shows that bigger TA population sizes tend to have higher theft rates. The scatter plot in Figure 4 shows an upward trend, as shown by the smoothed trend line fitted to the points. The trend line is flat on the left, rises through the middle and slightly flattens on the right. The shape of the points also shows an upward trend, and theft rates vary more as the TAs increase.

5. Detailed Analysis Results

5.1 Overview

This section moves beyond exploratory data analysis to examine the correlation between the two variables: theft rate and population. It further assesses model performance using k-fold cross-validation and discusses the associated limitations.

5.2 Data Transformation

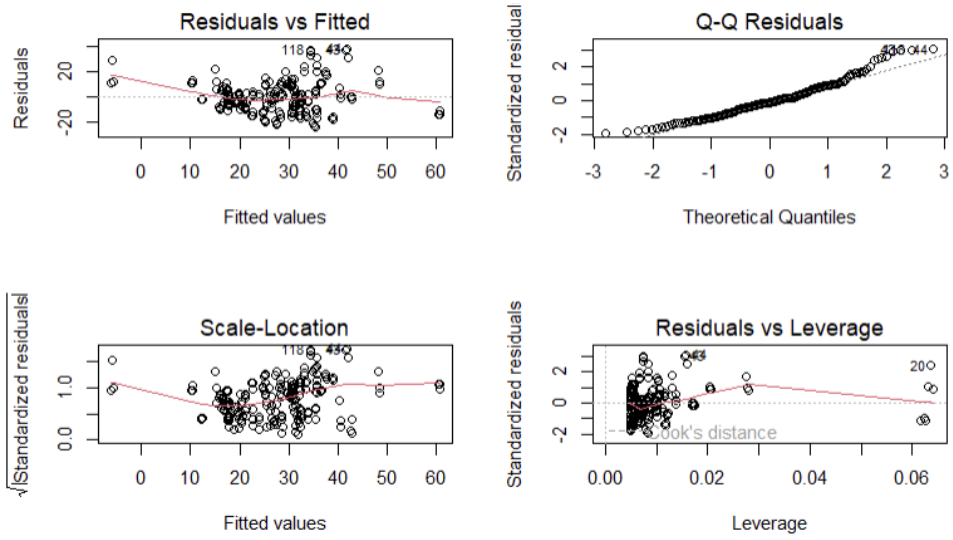


Figure 5. Diagnostic plots of the model, recorded theft rates against log(population).

Figure 5 shows diagnostic plots that allow us to check the assumptions of linearity, normality of residuals, homoscedasticity (constant variance), and non-influential points. The Residuals vs. Fitted plot displays signs of non-linearity. The Q-Q plot shows a deviation of the residuals from the reference line, indicating that the assumption of normality does not hold. The Scale-Location plot shows a slight curve, indicating that the homoscedasticity assumption may also be violated. On the Residuals vs. Leverage plot, there are no influential points shown, which is a positive finding.

However, because the other assumptions are violated, it signals to us that a log transformation of the theft rate is needed.

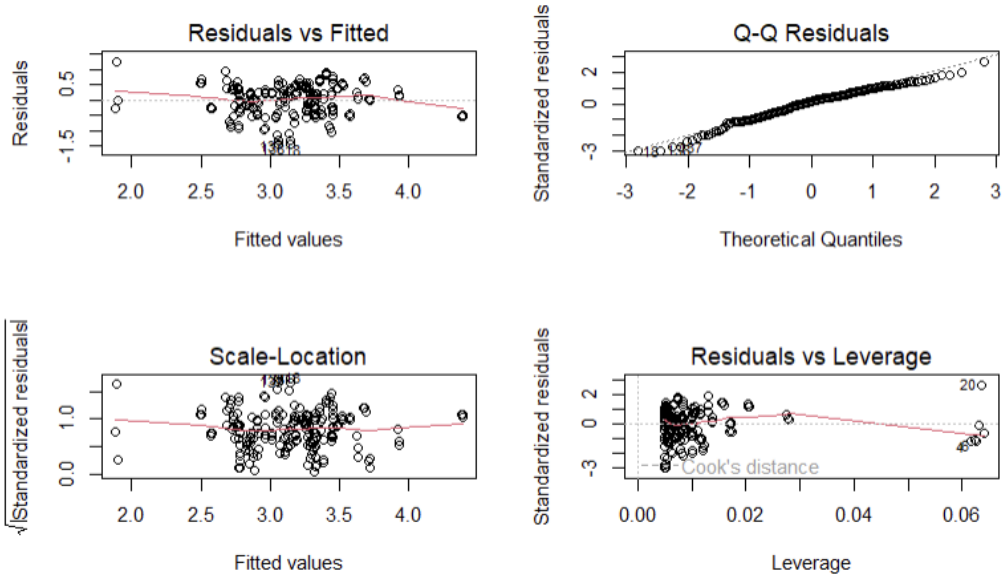


Figure 6. Diagnostic plots of the model, recorded log(theft_rates) against log(population).

Figure 6 shows the same diagnostic plots as Figure 5, but with a log transformation applied to the recorded theft rates (now both variables have a log transformation). The diagnostic plots for this new model show that it fits the data much more accurately than before. The Residuals vs. Fitted plot has improved, and the assumption of linearity holds. The points on the Q-Q plot follow the reference line more closely; although there is a slight deviation, the plot indicates that the assumption of normality holds. The Scale-Location plot shows constant variance, and like before, there are no influential points on the Residuals vs. Leverage plot. This demonstrates that the four linear regression assumptions are now satisfied, ensuring that our results and findings are reliable.

5.3 Correlation Results

Correlation Type	n	Coefficient	95% CI	p-value
Pooled Pearson	201	0.616	[0.52, 0.70]	<0.001
Pooled Spearman	201	0.626	-	<0.001
Spearman FY2023	67	0.653	-	<0.001
Spearman FY2024	67	0.603	-	<0.001
Spearman FY2025	67	0.629	-	<0.001

Table 2: Correlation values, p-values, and 95% confidence intervals (where applicable) for theft rate and population. Results include pooled data (Pearson and Spearman) as well as Spearman rank correlations across individual years.

Table 2 shows the correlation values for the recorded theft rate and population, using both Pearson and Spearman methods to assess their association. The pooled results yield a correlation coefficient of approximately 0.62, indicating a moderate positive relationship. However, a key limitation is that the 201 total observations consist of 67 Territorial Authorities (TAs) tracked across three financial years (FY2023, FY2024, and FY2025), meaning standard correlation methods treat them as independent points when they are not. To address this, the dataset was analysed by individual financial years. We observed consistent correlation values across all separate years, with a slight increase in FY2023(0.653), indicating that our findings remain stable over time. This consistency allows us to conclude that larger TA populations tend to have higher recorded theft rates.

5.4 Predictive Performance: K-fold Cross-validation

Model RMSE	Baseline RMSE	Difference	Standard Deviation
12.56	15.4	2.83	15.88

Table 3: K-fold cross-validation RMSE (Root Mean Square Error) comparison between the model and baseline, including performance difference and standard deviation.

Table 3 shows the predictive performance of the K-fold cross-validation (measured in recorded thefts per 1,000 residents). To assess its accuracy, a baseline model and standard deviation were included. The model achieved a K-fold CV value of 12.56 (seen in 'Model RMSE'), which represents the average magnitude of the prediction errors. The k-fold cv model achieved a lower RMSE value than the baseline, with an error reduction of 2.83 units between the cross-validation model and the baseline model, demonstrating that the model makes more accurate predictions. This shows us that population helps with predicting theft rate, but it also has a large RMSE rate with a lot of variation left unexplained ($R^2 = 0.363$), which shows us that it's not a strong predictor on its own.

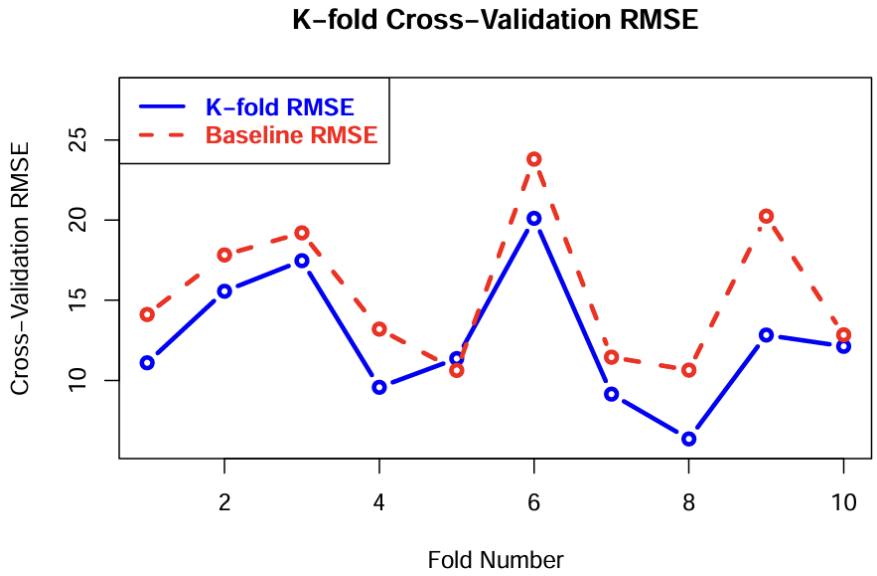


Figure 7: K-fold cross-validation RMSE (Root Mean Square Error) plot comparison between the model and baseline.

In addition, Figure 7 displays the RMSE (measured in recorded thefts per 1,000 residents) plotted across the 10 individual folds. This plot visually confirmed how the model (RMSE = 12.56) compares to the baseline (RMSE = 15.40) across different subsets of the data. Both lines track each other relatively closely, indicating that the model is stable. This plot showed us that the k-fold CV model was moderately lower than the baseline model. But at fold 5, the baseline model scored a slightly lower error rate. The sudden spikes and drops (folds 6 & 8) show us that our model is sensitive to data splits and which TA is selected in each fold, which reveals to us that the variability amongst the TAs is an issue. This demonstrates to us that while the models show consistency across the folds, the estimates remain sensitive to high variability among the TA.

6. Conclusions and Recommendations

Throughout this project, we were able to meaningfully extract insights from the NZ Police and Stats NZ data. I discovered that there was a moderate correlation between population and theft rate. This association doesn't imply a cause, but it reveals that bigger population sizes of TAs tend to have higher theft rates. I also explored the performance of predicting theft rate with TA population size. This covers our question, but our research had a few limitations that prevented its robustness.

The recorded victimisations we focused on in this project didn't account for crimes that were unreported. Dwelling crimes other than burglary were omitted from public data available on the NZ Police website, which means that our findings don't fully represent victimisations in NZ. This means it isn't suitable for us to completely trust the findings and make crucial decisions from this research unconditionally. This research allowed us to understand the general trend of victimisations across TAs in New Zealand, with considerations.

I achieved an adjusted R^2 of 0.363, which only explains roughly a third of the variation in our log-transformed model with population. This is also a limitation for our project and indicates that using one predictor on its own to predict theft rate is not recommended. It would be best to add more predictors to our model to attain more accurate predictions.

For the reader, this means that they should take these insights with moderation. Stakeholders such as residents, business owners and government officials will be able to use this as a directional guide for their research. This research allowed us to display how data science was able to address the research question with evidence to support it.

For more rigorous research, I would suggest further research to be done as to which predictors should be selected, such as a 'best subset selection', for the model to train on. I only covered the financial years 2023, 2024 and 2025. With more years to train on, the model will have more data to learn from and will perform more accurately in its predictions.

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